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Otomasi Tugas dengan Python

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Apa itu Otomasi?

Otomasi adalah proses membuat tugas yang biasanya dilakukan manual menjadi otomatis menggunakan program.

Contoh Otomasi di Kehidupan Sehari-hari:

🔔 Alarm HP (otomatis bunyi di waktu tertentu)
🌡️ AC otomatis mati jika suhu sudah dingin
💡 Lampu sensor gerak
🚿 Keran otomatis di toilet

Tugas yang Bisa Diotomasi dengan Python:

📁 Mengelola file (rename, move, backup)
📊 Mengolah data Excel otomatis
📧 Mengirim email massal
📸 Mengedit banyak foto sekaligus
📝 Generate laporan otomatis
🔄 Backup data berkala

Otomasi Pengelolaan File

A. Organisasi File Otomatis :

- Folder Download berantakan
- File tidak terorganisir
- Sulit cari file

```python
import os
import shutil

def organize_files(folder_path):
    """
    Organisir file berdasarkan ekstensi
    """

    # Definisi kategori file
    file_types = {
        'Images': ['.jpg', '.jpeg', '.png', '.gif', '.bmp', '.svg'],
        'Documents': ['.pdf', '.doc', '.docx', '.txt', '.ppt', '.pptx', '.xls', '.xlsx'],
        'Videos': ['.mp4', '.avi', '.mkv', '.mov', '.flv'],
        'Music': ['.mp3', '.wav', '.flac', '.m4a'],
        'Archives': ['.zip', '.rar', '.7z', '.tar', '.gz'],
        'Programs': ['.exe', '.msi', '.apk'],
        'Code': ['.py', '.java', '.cpp', '.js', '.html', '.css']
    }

    print(" Mulai mengorganisir file...")

    # Loop semua file di folder
    for filename in os.listdir(folder_path):
        file_path = os.path.join(folder_path, filename)

        # Skip jika folder
        if os.path.isdir(file_path):
            continue

        # Ambil ekstensi file
        file_ext = os.path.splitext(filename)[1].lower()

        # Cari kategori yang sesuai
        moved = False
        for category, extensions in file_types.items():
            if file_ext in extensions:
                # Buat folder kategori jika belum ada
                category_path = os.path.join(folder_path, category)
                if not os.path.exists(category_path):
                    os.makedirs(category_path)

                # Pindahkan file
                destination = os.path.join(category_path, filename)

                # Jika file sudah ada, tambahkan angka
                if os.path.exists(destination):
                    base, ext = os.path.splitext(filename)
                    counter = 1
                    while os.path.exists(destination):
                        new_filename = f"{base}_{counter}{ext}"
                        destination = os.path.join(category_path, new_filename)
                        counter += 1

                shutil.move(file_path, destination)
                print(f"{filename} → {category}/")
                moved = True
                break

        # Jika tidak ada kategori yang cocok, pindah ke "Others"
        if not moved:
            others_path = os.path.join(folder_path, 'Others')
            if not os.path.exists(others_path):
                os.makedirs(others_path)

            destination = os.path.join(others_path, filename)
            shutil.move(file_path, destination)
            print(f"📦 {filename} → Others/")

    print("\n✨ Selesai! File sudah terorganisir.")

# Cara pakai
if __name__ == "__main__":
    # Ganti dengan path folder yang ingin diorganisir
    folder = r"C:\Users\YourName\Downloads"  # Windows
    # folder = "/Users/YourName/Downloads"    # Mac/Linux

    organize_files(folder)
```

Otomasi Pengelolaan File

B. Backup File Otomatis

```python
import os
import shutil
from datetime import datetime

def backup_folder(source, backup_location):
    """
    Backup folder dengan timestamp
    """

    # Buat nama backup dengan tanggal
    timestamp = datetime.now().strftime("%Y-%m-%d_%H-%M-%S")
    folder_name = os.path.basename(source)
    backup_name = f"{folder_name}_backup_{timestamp}"
    backup_path = os.path.join(backup_location, backup_name)

    try:
        print(f"📦 Membackup: {source}")
        print(f"📁 Ke: {backup_path}")

        # Copy seluruh folder
        shutil.copytree(source, backup_path)

        # Hitung ukuran backup
        total_size = 0
        for dirpath, dirnames, filenames in os.walk(backup_path):
            for filename in filenames:
                filepath = os.path.join(dirpath, filename)
                total_size += os.path.getsize(filepath)

        size_mb = total_size / (1024 * 1024)

        print(f"✅ Backup berhasil!")
        print(f"📊 Ukuran: {size_mb:.2f} MB")
        print(f"📅 Waktu: {timestamp}")

        return backup_path

    except Exception as e:
        print(f"❌ Error: {e}")
        return None

# Cara pakai
if __name__ == "__main__":
    source_folder = r"C:\Users\YourName\Documents\Tugas"
    backup_location = r"D:\Backup"

    backup_folder(source_folder, backup_location)
```

Otomasi Pengelolaan File

C. Rename File Massal

```python
import os

def batch_rename(folder_path, prefix="", start_number=1):
    """
    Rename banyak file sekaligus dengan format:
    prefix_001.ext, prefix_002.ext, dst.
    """

    files = [f for f in os.listdir(folder_path) if os.path.isfile(os.path.join(folder_path, f))]
    files.sort()

    print(f"📁 Ditemukan {len(files)} file")
    print(f"🔄 Mulai rename dengan prefix: '{prefix}'\n")

    for index, filename in enumerate(files, start=start_number):
        # Ambil ekstensi file
        _, ext = os.path.splitext(filename)

        # Buat nama baru
        new_name = f"{prefix}_{index:03d}{ext}"

        # Path lengkap
        old_path = os.path.join(folder_path, filename)
        new_path = os.path.join(folder_path, new_name)

        # Rename
        os.rename(old_path, new_path)
        print(f"✅ {filename} → {new_name}")

    print(f"\n✨ Selesai! {len(files)} file berhasil direname.")

# Cara pakai
if __name__ == "__main__":
    folder = r"C:\Users\YourName\Pictures\Vacation"
    batch_rename(folder, prefix="Liburan_Bali", start_number=1)
```

Otomasi Email

Instalasi Library:

```bash
pip install secure-smtplib
# Atau gunakan library bawaan Python
```

Instalasi Library:

```python
import smtplib
from email.mime.text import MIMEText
from email.mime.multipart import MIMEMultipart
from email.mime.base import MIMEBase
from email import encoders

def send_email(sender_email, sender_password, recipient_email, subject, body, attachment=None):
    """
    Kirim email dengan atau tanpa attachment
    """

    try:
        # Setup email
        msg = MIMEMultipart()
        msg['From'] = sender_email
        msg['To'] = recipient_email
        msg['Subject'] = subject

        # Body email
        msg.attach(MIMEText(body, 'plain'))

        # Attachment (jika ada)
        if attachment:
            with open(attachment, 'rb') as file:
                part = MIMEBase('application', 'octet-stream')
                part.set_payload(file.read())
                encoders.encode_base64(part)
                part.add_header('Content-Disposition', f'attachment; filename= {attachment}')
                msg.attach(part)

        # Koneksi ke Gmail SMTP
        server = smtplib.SMTP('smtp.gmail.com', 587)
        server.starttls()
        server.login(sender_email, sender_password)

        # Kirim email
        text = msg.as_string()
        server.sendmail(sender_email, recipient_email, text)
        server.quit()

        print(f"✅ Email berhasil dikirim ke {recipient_email}")
        return True

    except Exception as e:
        print(f"❌ Error: {e}")
        return False

# Cara pakai
if __name__ == "__main__":
    sender = "your_email@gmail.com"
    password = "your_app_password"  # Gunakan App Password, bukan password biasa!
    recipient = "friend@gmail.com"

    subject = "Laporan Tugas Kelompok"
    body = """
    Hai Tim,

    Terlampir adalah laporan tugas kelompok kita.
    Mohon dicek dan beri feedback.

    Terima kasih!
    """

    attachment = "laporan.pdf"

    send_email(sender, password, recipient, subject, body, attachment)
```

Otomasi Email

Kirim Email Massal:

```python
import pandas as pd
import time

def send_bulk_emails(sender_email, sender_password, csv_file):
    """
    Kirim email ke banyak penerima dari file CSV

    CSV format:
    nama,email,nilai
    Ali,ali@email.com,85
    Budi,budi@email.com,90
    """

    # Baca data dari CSV
    df = pd.read_csv(csv_file)

    print(f"📧 Akan mengirim email ke {len(df)} penerima\n")

    success = 0
    failed = 0

    for index, row in df.iterrows():
        nama = row['nama']
        email = row['email']
        nilai = row['nilai']

        # Personalisasi subject dan body
        subject = f"Nilai Ujian - {nama}"
        body = f"""
        Halo {nama},

        Berikut adalah nilai ujian kamu:
        Nilai: {nilai}

        {"Selamat! Nilai kamu bagus!" if nilai >= 75 else "Tetap semangat! Belajar lebih giat lagi."}

        Salam,
        Guru
        """

        print(f"📤 Mengirim ke {nama} ({email})...")

        if send_email(sender_email, sender_password, email, subject, body):
            success += 1
        else:
            failed += 1

        # Jeda agar tidak dianggap spam
        time.sleep(2)

    print(f"\n📊 Statistik:")
    print(f"✅ Berhasil: {success}")
    print(f"❌ Gagal: {failed}")

# Cara pakai
if __name__ == "__main__":
    sender = "guru@school.com"
    password = "app_password"
    csv_file = "daftar_nilai.csv"

    send_bulk_emails(sender, password, csv_file)
```

Otomasi Excel dengan openpyxl

Install : pip install openpyxl

A. Generate Laporan Otomatis:

```python
from openpyxl import Workbook
from openpyxl.styles import Font, Alignment, PatternFill
from openpyxl.chart import BarChart, Reference
from datetime import datetime

def create_report(data, filename):
    """
    Buat laporan Excel otomatis dengan format dan chart

    data = {
        'Nama': ['Ali', 'Budi', 'Citra'],
        'Matematika': [85, 78, 92],
        'Fisika': [80, 85, 88],
        'Kimia': [88, 76, 90]
    }
    """

    wb = Workbook()
    ws = wb.active
    ws.title = "Laporan Nilai"

    # Header
    ws['A1'] = "LAPORAN NILAI SISWA"
    ws['A1'].font = Font(size=16, bold=True, color="FFFFFF")
    ws['A1'].fill = PatternFill(start_color="4472C4", end_color="4472C4", fill_type="solid")
    ws['A1'].alignment = Alignment(horizontal="center", vertical="center")
    ws.merge_cells('A1:E1')
    ws['A2'] = f"Tanggal: {datetime.now().strftime('%d-%m-%Y')}"
    ws['A2'].alignment = Alignment(horizontal="center")
    ws.merge_cells('A2:E2')

    # Header tabel (baris 4)
    headers = ['No', 'Nama', 'Matematika', 'Fisika', 'Kimia', 'Rata-rata']
    for col, header in enumerate(headers, start=1):
        cell = ws.cell(row=4, column=col, value=header)
        cell.font = Font(bold=True, color="FFFFFF")
        cell.fill = PatternFill(start_color="70AD47", end_color="70AD47", fill_type="solid")
        cell.alignment = Alignment(horizontal="center")

    # Data siswa
    nama_list = data['Nama']
    for i, nama in enumerate(nama_list, start=1):
        row = i + 4  # Mulai dari baris 5

        ws.cell(row=row, column=1, value=i)
        ws.cell(row=row, column=2, value=nama)
        ws.cell(row=row, column=3, value=data['Matematika'][i-1])
        ws.cell(row=row, column=4, value=data['Fisika'][i-1])
        ws.cell(row=row, column=5, value=data['Kimia'][i-1])

        # Rata-rata dengan formula
        ws.cell(row=row, column=6, value=f"=AVERAGE(C{row}:E{row})")
        ws.cell(row=row, column=6).number_format = '0.00'
```

Otomasi Excel dengan openpyxl

A. Generate Laporan Otomatis:

```python
    # Tambah baris total rata-rata
    last_row = len(nama_list) + 5
    ws.cell(row=last_row, column=1, value="RATA-RATA KELAS")
    ws.cell(row=last_row, column=1).font = Font(bold=True)
    ws.merge_cells(f'A{last_row}:B{last_row}')

    for col in range(3, 7):
        ws.cell(row=last_row, column=col, value=f"=AVERAGE({chr(64+col)}5:{chr(64+col)}{last_row-1})")
        ws.cell(row=last_row, column=col).number_format = '0.00'
        ws.cell(row=last_row, column=col).font = Font(bold=True)
        ws.cell(row=last_row, column=col).fill = PatternFill(start_color="FFC000", end_color="FFC000", fill_type="solid")

    # Atur lebar kolom
    ws.column_dimensions['A'].width = 5
    ws.column_dimensions['B'].width = 20
    ws.column_dimensions['C'].width = 12
    ws.column_dimensions['D'].width = 12
    ws.column_dimensions['E'].width = 12
    ws.column_dimensions['F'].width = 12

    # Buat chart
    chart = BarChart()
    chart.title = "Grafik Nilai Siswa"
    chart.x_axis.title = "Siswa"
    chart.y_axis.title = "Nilai"
```

```python
    # Data untuk chart
    data_ref = Reference(ws, min_col=3, min_row=4, max_col=5, max_row=last_row-1)
    categories = Reference(ws, min_col=2, min_row=5, max_row=last_row-1)

    chart.add_data(data_ref, titles_from_data=True)
    chart.set_categories(categories)

    # Posisi chart
    ws.add_chart(chart, f"H4")

    # Save
    wb.save(filename)
    print(f"✅ Laporan berhasil dibuat: {filename}")

# Cara pakai
if __name__ == "__main__":
    data_siswa = {
        'Nama': ['Ali Rahman', 'Budi Santoso', 'Citra Dewi', 'Dina Maharani', 'Eko Prasetyo'],
        'Matematika': [85, 78, 92, 88, 75],
        'Fisika': [80, 85, 89, 82, 78],
        'Kimia': [88, 76, 94, 85, 80]
    }

    create_report(data_siswa, "Laporan_Nilai.xlsx")
```

📊 Otomasi Excel dengan openpyxl

B. Membaca dan Mengolah Data Excel:

```python
from openpyxl import load_workbook
import statistics

def analyze_excel(filename):
    """
    Analisis data dari Excel
    """

    wb = load_workbook(filename)
    ws = wb.active

    print("📊 ANALISIS DATA NILAI")
    print("="*50)

    # Baca data (skip header)
    data = []
    for row in ws.iter_rows(min_row=5, max_row=ws.max_row-1, min_col=3, max_col=5, values_only=True):
        if all(isinstance(val, (int, float)) for val in row):
            data.append(row)

    if not data:
        print("❌ Tidak ada data untuk dianalisis")
        return

    # Analisis per mata pelajaran
    mapel = ['Matematika', 'Fisika', 'Kimia']

    for i, nama_mapel in enumerate(mapel):
        nilai_mapel = [row[i] for row in data]

        print(f"\n📖 {nama_mapel}:")
        print(f"   Rata-rata: {statistics.mean(nilai_mapel):.2f}")
        print(f"   Tertinggi: {max(nilai_mapel)}")
        print(f"   Terendah: {min(nilai_mapel)}")
        print(f"   Median: {statistics.median(nilai_mapel):.2f}")

        # Hitung siswa lulus (≥75)
        lulus = sum(1 for n in nilai_mapel if n >= 75)
        persen_lulus = (lulus / len(nilai_mapel)) * 100
        print(f"   Lulus: {lulus}/{len(nilai_mapel)} ({persen_lulus:.1f}%)")

# Cara pakai
analyze_excel("Laporan_Nilai.xlsx")
```

🖼️ Otomasi Pengolahan Gambar

Install : pip install Pillow

A. Resize Banyak Foto Sekaligus:

```python
from PIL import Image
import os

def batch_resize(input_folder, output_folder, width=800, height=600):
    """
    Resize banyak foto sekaligus
    """

    # Buat folder output jika belum ada
    if not os.path.exists(output_folder):
        os.makedirs(output_folder)

    # Format gambar yang didukung
    formats = ('.jpg', '.jpeg', '.png', '.bmp', '.gif')

    files = [f for f in os.listdir(input_folder) if f.lower().endswith(formats)]

    print(f"🖼  Ditemukan {len(files)} gambar")
    print(f"📏 Ukuran target: {width}x{height}px\n")

    for filename in files:
        input_path = os.path.join(input_folder, filename)
        output_path = os.path.join(output_folder, filename)

        try:
            # Buka gambar
            img = Image.open(input_path)

            # Resize (maintain aspect ratio)
            img.thumbnail((width, height), Image.Resampling.LANCZOS)

            # Save
            img.save(output_path, quality=90, optimize=True)

            print(f"✅ {filename} → {img.size[0]}x{img.size[1]}px")

        except Exception as e:
            print(f"❌ Error pada {filename}: {e}")

    print(f"\n✨ Selesai! Hasil disimpan di: {output_folder}")

# Cara pakai
if __name__ == "__main__":
    input_folder = r"C:\Users\YourName\Pictures\Original"
    output_folder = r"C:\Users\YourName\Pictures\Resized"

    batch_resize(input_folder, output_folder, width=1920, height=1080)
```

🖼️ Otomasi Pengolahan Gambar

B. Tambah Watermark Otomatis:

```python
from PIL import Image, ImageDraw, ImageFont
import os

def add_watermark(input_folder, output_folder, watermark_text="© 2026"):
    """
    Tambah watermark ke banyak foto
    """

    if not os.path.exists(output_folder):
        os.makedirs(output_folder)

    formats = ('.jpg', '.jpeg', '.png')
    files = [f for f in os.listdir(input_folder) if f.lower().endswith(formats)]

    print(f"🖼  Menambah watermark ke {len(files)} gambar")
    print(f"©  Watermark: '{watermark_text}'\n")

    for filename in files:
        input_path = os.path.join(input_folder, filename)
        output_path = os.path.join(output_folder, filename)

        try:
            # Buka gambar
            img = Image.open(input_path)
            width, height = img.size

            # Buat layer untuk watermark
            draw = ImageDraw.Draw(img)

            # Font (gunakan font default jika tidak ada)
            try:
                font = ImageFont.truetype("arial.ttf", int(width * 0.03))
            except:
                font = ImageFont.load_default()
```

```python
            # Hitung posisi (kanan bawah)
            text_bbox = draw.textbbox((0, 0), watermark_text, font=font)
            text_width = text_bbox[2] - text_bbox[0]
            text_height = text_bbox[3] - text_bbox[1]

            margin = 20
            x = width - text_width - margin
            y = height - text_height - margin

            # Tambahkan shadow
            shadow_offset = 2
            draw.text((x+shadow_offset, y+shadow_offset), watermark_text,
                      font=font, fill=(0, 0, 0, 128))

            # Tambahkan watermark
            draw.text((x, y), watermark_text, font=font, fill=(255, 255, 255, 200))

            # Save
            img.save(output_path, quality=95)
            print(f"✅ {filename}")

        except Exception as e:
            print(f"❌ Error pada {filename}: {e}")

    print(f"\n✨ Selesai! Hasil disimpan di: {output_folder}")

# Cara pakai
input_folder = r"C:\Users\YourName\Pictures\Photos"
output_folder = r"C:\Users\YourName\Pictures\Watermarked"

add_watermark(input_folder, output_folder, watermark_text="© SMA Ku 2026")
```

17 Scheduler - Jalankan Program Otomatis

Install : pip install schedule

Contoh: Backup Otomatis Setiap Hari:

```python
import schedule
import time
import shutil
from datetime import datetime
import os

def backup_task():
    """Tugas backup yang akan dijalankan otomatis"""

    source = r"C:\Users\YourName\Documents\Tugas"
    backup_location = r"D:\Backup"

    timestamp = datetime.now().strftime("%Y-%m-%d_%H-%M")
    backup_name = f"Tugas_Backup_{timestamp}"
    backup_path = os.path.join(backup_location, backup_name)

    try:
        shutil.copytree(source, backup_path)
        print(f"✅ [{timestamp}] Backup berhasil: {backup_path}")
    except Exception as e:
        print(f"❌ [{timestamp}] Backup gagal: {e}")

# Jadwalkan tugas
schedule.every().day.at("20:00").do(backup_task)  # Setiap hari jam 20:00
# schedule.every().hour.do(backup_task)           # Setiap jam
# schedule.every(30).minutes.do(backup_task)      # Setiap 30 menit
# schedule.every().monday.at("09:00").do(backup_task)  # Setiap Senin jam 09:00

print("⏰ Scheduler aktif!")
print("📅 Backup otomatis setiap hari jam 20:00")
print("⏸️  Tekan Ctrl+C untuk stop\n")

# Loop utama
while True:
    schedule.run_pending()
    time.sleep(60)  # Cek setiap 1 menit
```

🎮 Proyek Lengkap: Aplikasi Auto-Organizer

```python
"""
AUTO-ORGANIZER
Aplikasi untuk mengorganisir file, backup, dan reminder
"""

import os
import shutil
from datetime import datetime
import schedule
import time

class AutoOrganizer:
    def __init__(self):
        self.config = {
            'watch_folder': r"C:\Users\YourName\Downloads",
            'backup_location': r"D:\Backup",
            'organize_time': "21:00",  # Jam 9 malam
            'backup_time': "22:00"      # Jam 10 malam
        }

        self.file_types = {
            'Images': ['.jpg', '.jpeg', '.png', '.gif', '.bmp'],
            'Documents': ['.pdf', '.doc', '.docx', '.txt', '.ppt', '.xlsx'],
            'Videos': ['.mp4', '.avi', '.mkv', '.mov'],
            'Music': ['.mp3', '.wav', '.flac'],
            'Archives': ['.zip', '.rar', '.7z'],
            'Code': ['.py', '.java', '.cpp', '.js', '.html', '.css']
        }
```

```python
    def organize_files(self):
        """Organisir file"""
        folder = self.config['watch_folder']
        timestamp = datetime.now().strftime("%H:%M:%S")

        print(f"\n[{timestamp}] 🔄 Mulai mengorganisir file...")

        count = 0
        for filename in os.listdir(folder):
            file_path = os.path.join(folder, filename)

            if os.path.isdir(file_path):
                continue

            file_ext = os.path.splitext(filename)[1].lower()

            for category, extensions in self.file_types.items():
                if file_ext in extensions:
                    category_path = os.path.join(folder, category)
                    if not os.path.exists(category_path):
                        os.makedirs(category_path)

                    destination = os.path.join(category_path, filename)

                    # Handle duplicate
                    if os.path.exists(destination):
                        base, ext = os.path.splitext(filename)
                        counter = 1
                        while os.path.exists
```

Thank *You!